1. Script to Find World-Writable Files

Answer:

A world-writable file means **others (everyone) have write permission** (`w`).
In Linux, this is dangerous because anyone can modify the file.

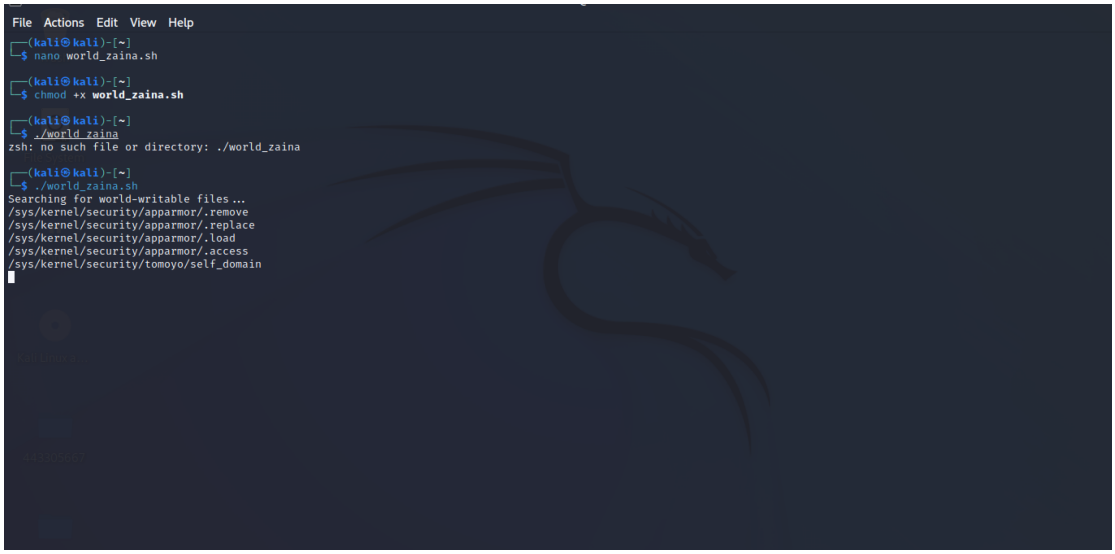
Bash Script:

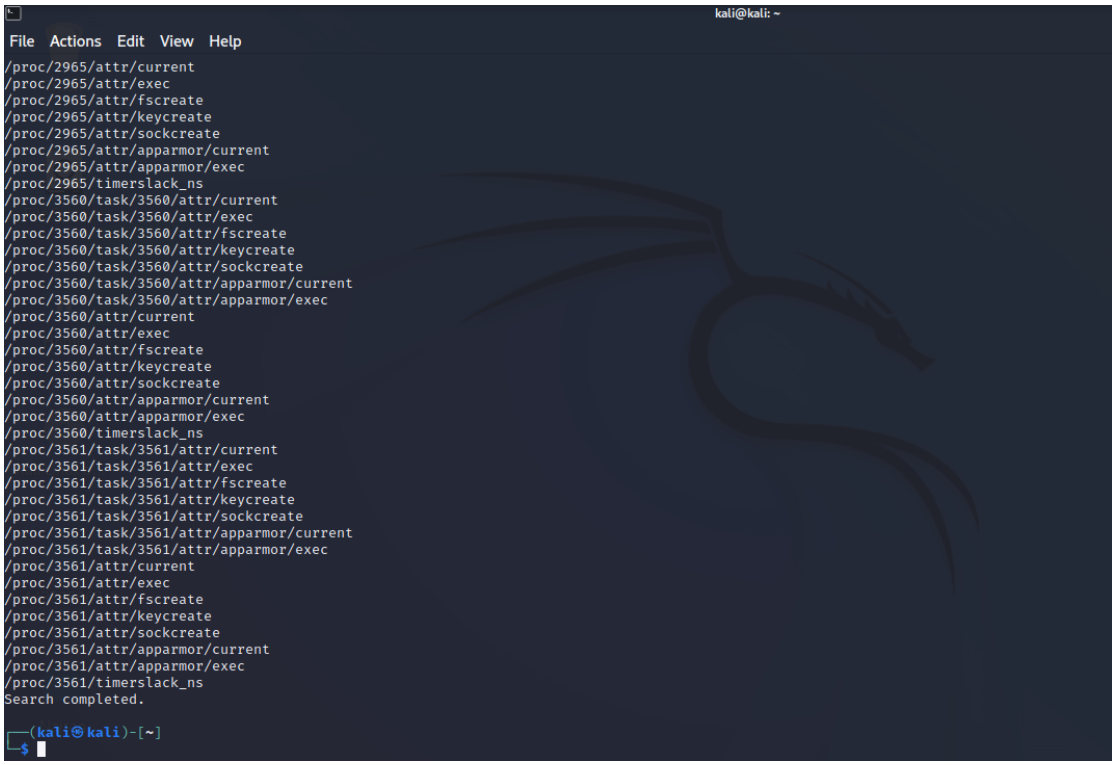
```
#!/bin/bash

echo "Searching for world-writable files..."

# Search from root directory
find / -type f -perm -002 2>/dev/null

echo "Search completed."
```





Explanation

- `find /` → starts searching from the root directory
- `-type f` → looks for files only
- `-perm -002` → finds files where **others have write permission**
- `2>/dev/null` → hides permission denied errors

2. How Windows NTFS Implements DAC

Answer:

Windows NTFS uses **Discretionary Access Control (DAC)** through something called **Access Control Lists (ACLs)**.

Explanation:

- Every file or folder in NTFS has a **security descriptor**
- This descriptor contains an **Access Control List (ACL)**
- The ACL includes multiple **Access Control Entries (ACEs)**

How it works:

- Each ACE defines:
    - A user or group
    - Their permissions (read, write, execute, full control)

Example:

- User A → Read only
- User B → Full control

Key Idea:

Just like Linux DAC:

- The **owner of the file decides who can access it**
- Permissions can be **granted or removed anytime**

Features of NTFS DAC:

- Supports users and groups
- Fine-grained permissions (more detailed than Linux)
- Permissions inheritance from parent folders

Limitation:

- If permissions are set incorrectly → security risk
- Similar to Linux chmod 777 problem